

TrES-3b

R-2790

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_120915 · created 2026-09-04T00:52:14+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2456185.68409 +/- 0.00096 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.133 +/- 0.016
Transit depth (Rp/Rs)^2	1.76 +/- 0.44 [%]
Orbital Inclination (inc)	86.99 +/- 0.93
Airmass coefficient 1 (a1)	1.194 +/- 0.015
Airmass coefficient 2 (a2)	-0.129 +/- 0.011
Scatter in the residuals of the lightcurve fit is	1.29 %
Best Comparison Star	None
Optimal Aperture	3.38
Optimal Annulus	8.31
Transit Duration (day)	0.0784 +/- 0.0027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.133 ± 0.016 vs 0.16309 (deviation 1.88σ)
Duration fit vs published	0.0784 d vs 0.0567 d (deviation 8.05σ)
Mid-time O–C	1.8 min
Depth SNR	8.3
Residual scatter	1.29 %

Light curve

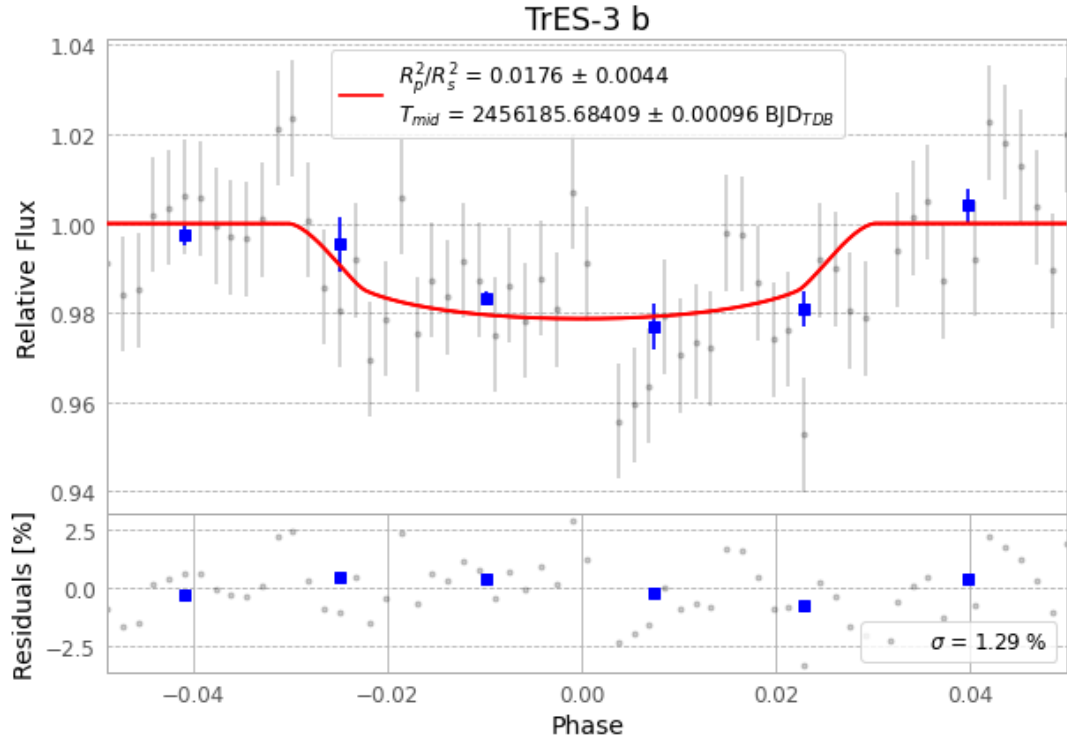


Figure 1 — FinalLightCurve_TrES-3 b_2012-09-14.png (SHA-256 993f547d94c7302f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [260, 296], 2 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c1ef4fa4961bc753...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

63 FITS frames (41 MB), TRES-3120915025112.FITS → TRES-3120915055713.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-120915.log (617300f5c90a...), FinalLightCurve_TrES-3 b_2012-09-14.png (993f547d94c7...), FinalLightCurve_TrES-3 b_2012-09-14.csv (c05987a95960...)

Compute resources

Wall time 126.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-04 00:52Z.